

Logika Fuzzy

Crisp Logic

- *Crisp logic is concerned with absolutes-true or false, there is no in-between.*

- Contoh:

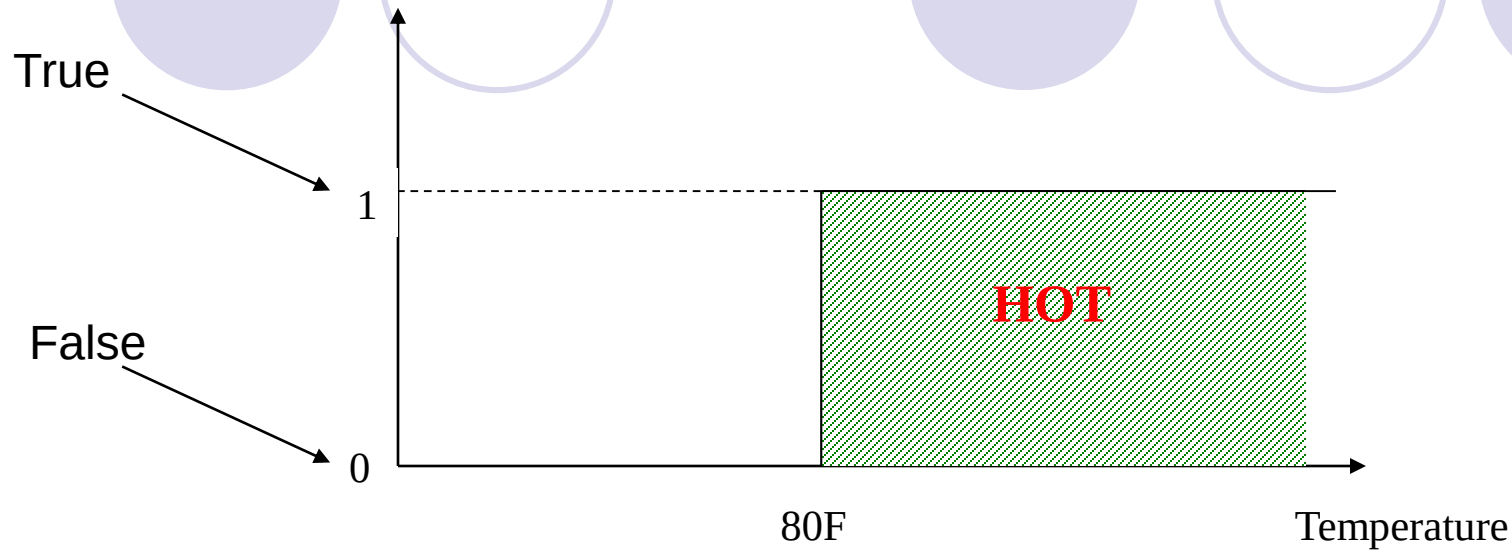
Rule:

If the temperature is higher than 80F, it is hot; otherwise, it is not hot.

Kasus:

- Temperature = 100F **Hot**
- Temperature = 80.1F **Hot**
- Temperature = 79.9F **Not hot**
- Temperature = 50F **Not hot**

Fungsi Keanggotaan untuk *crisp logic*

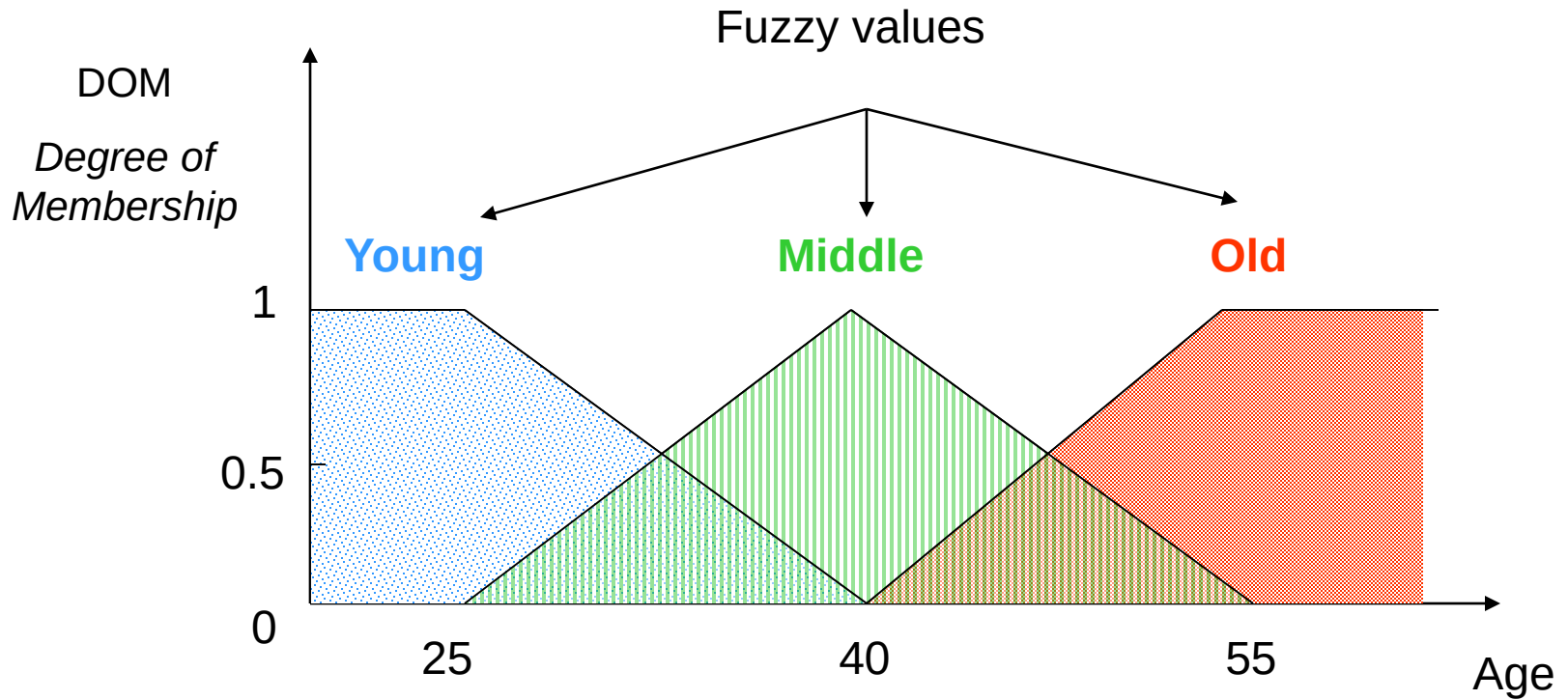


If temperature $\geq 80F$, it is hot (1 or true);

If temperature $< 80F$, it is not hot (0 or false).

- Fungsi keanggotaan dari crisp logic gagal membedakan antar member pada himpunan yang sama
- Ada problem-problem yang terlalu kompleks untuk didefinisikan secara tepat

Fungsi Keanggotaan Logika Fuzzy

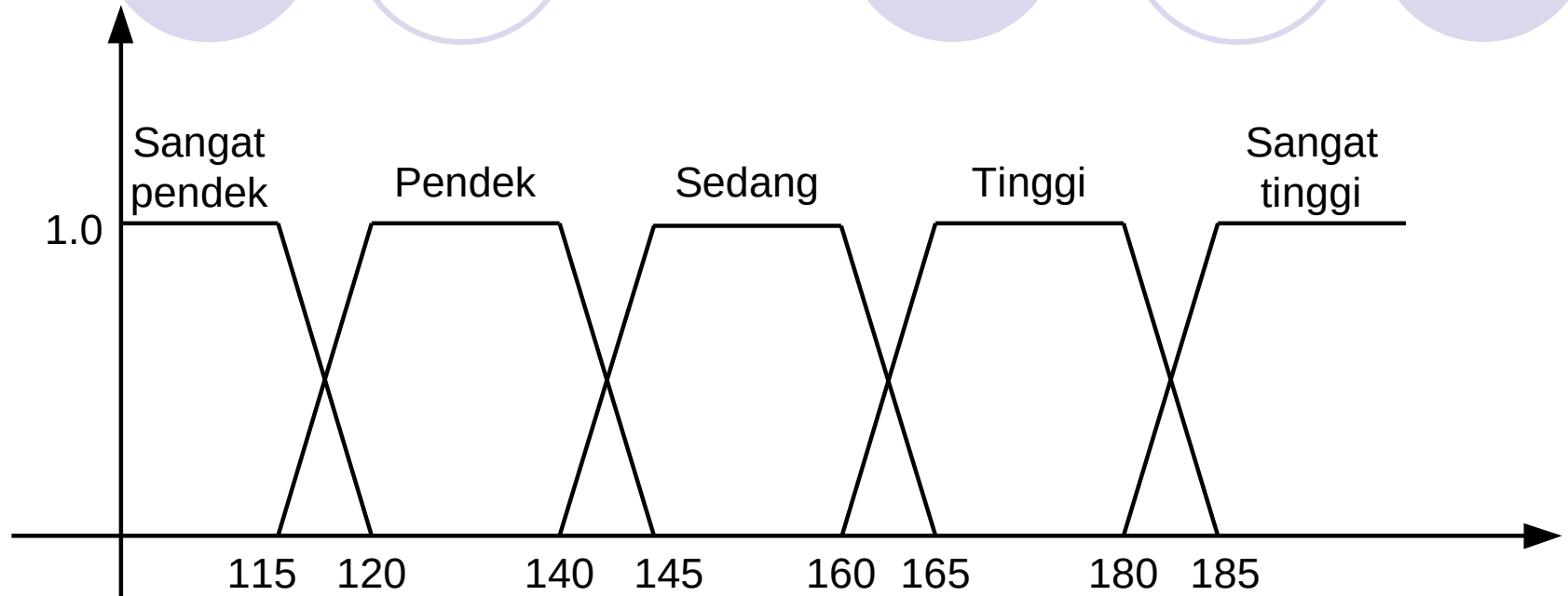


Nilai Fuzzy berasosiasi dengan derajat keanggotaan pada himpunan

Fuzzy vs Probabilitas

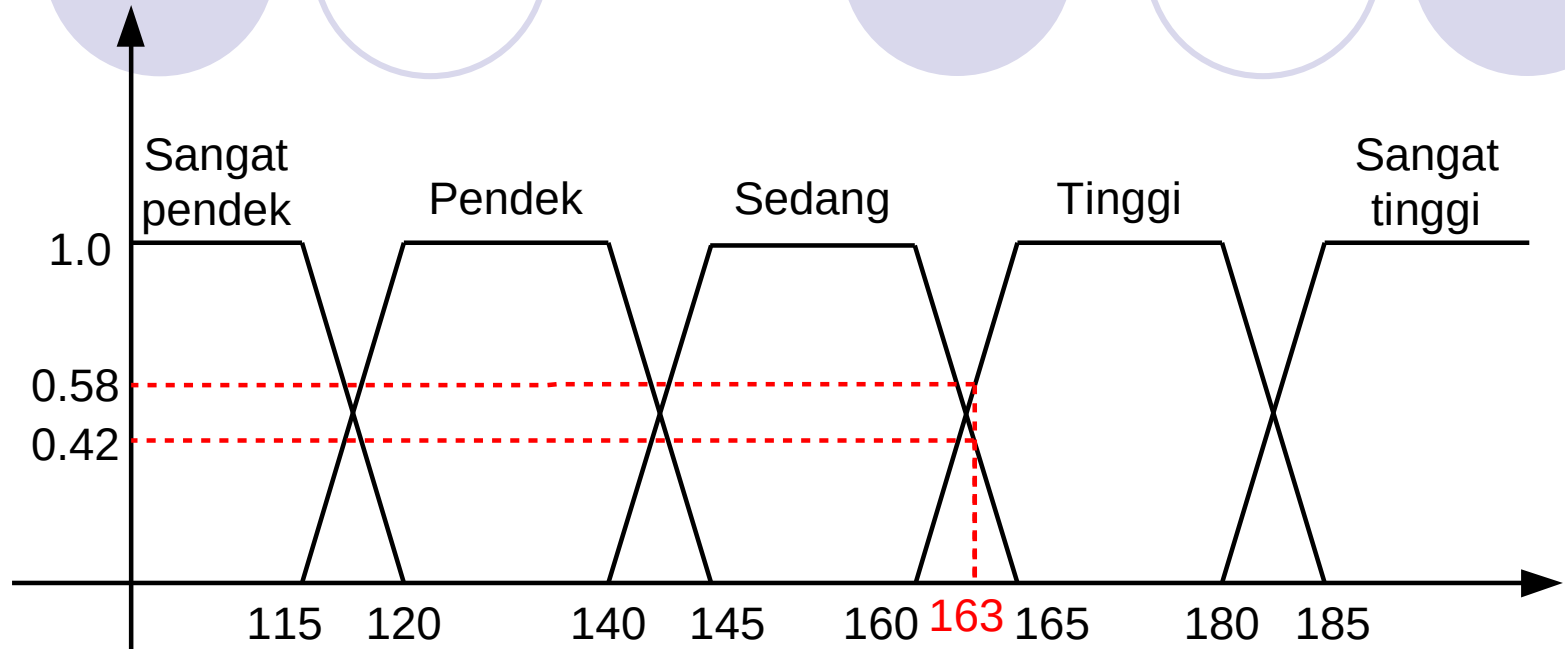
- Fuzzy $\neq$ Probabilitas
- - Probabilitas berkaitan dengan ketidakmenentuan dan kemungkinan
- - Logika Fuzzy berkaitan dengan ambiguitas dan ketidakjelasan
- Contoh 1:
Billy memiliki 10 jari kaki. Probabilitas Billy memiliki 9 jari kaki adalah 0. Keanggotaan Fuzzy Billy pada himpunan orang dengan 9 jari kaki $\neq 0$
- Contoh 2:
 - Probabilitas botol 1 berisi air beracun adalah 0.5 dan 0.5 untuk isi air murni { **mungkin air tersebut tidak beracun** }
 - Isi botol 2 memiliki nilai keanggotaan 0.5 pada himpunan air berisi racun { **air pasti beracun** }

Fuzzyfication (1)



$$\mu = [\mu_{sp}, \mu_p, \mu_s, \mu_t, \mu_{st}]$$

Fuzzyfication (2)

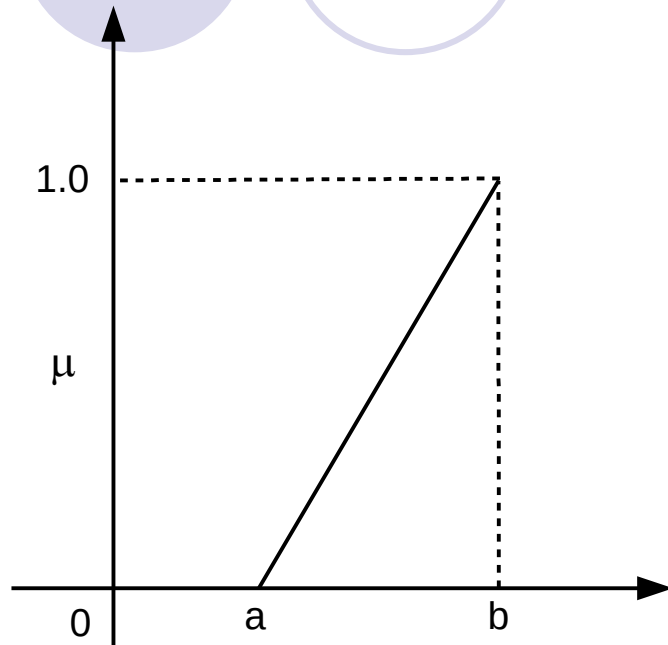


$$\mu[163] = [0, 0, 0.42, 0.58, 0]$$

atau

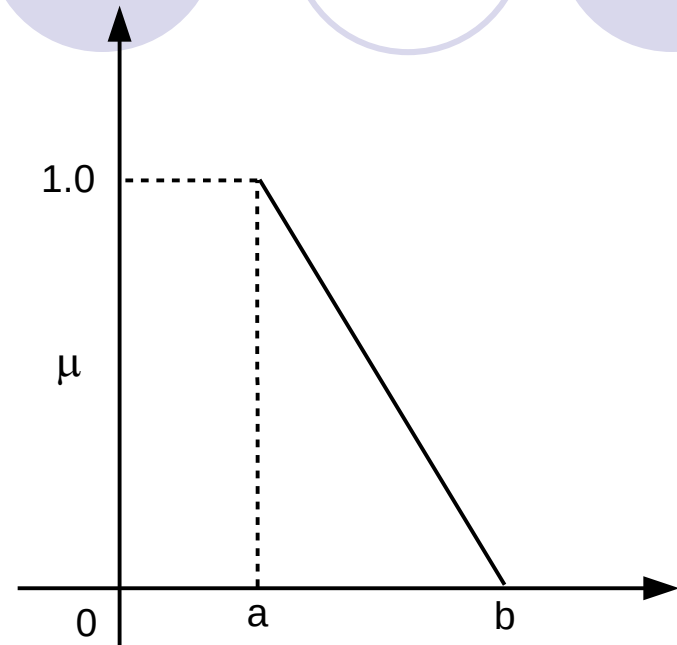
$$\mu_{\text{sedang}}[163] = 0.42, \mu_{\text{tinggi}}[163] = 0.58$$

Fungsi Keanggotaan: Fungsi Linier



Linier Naik

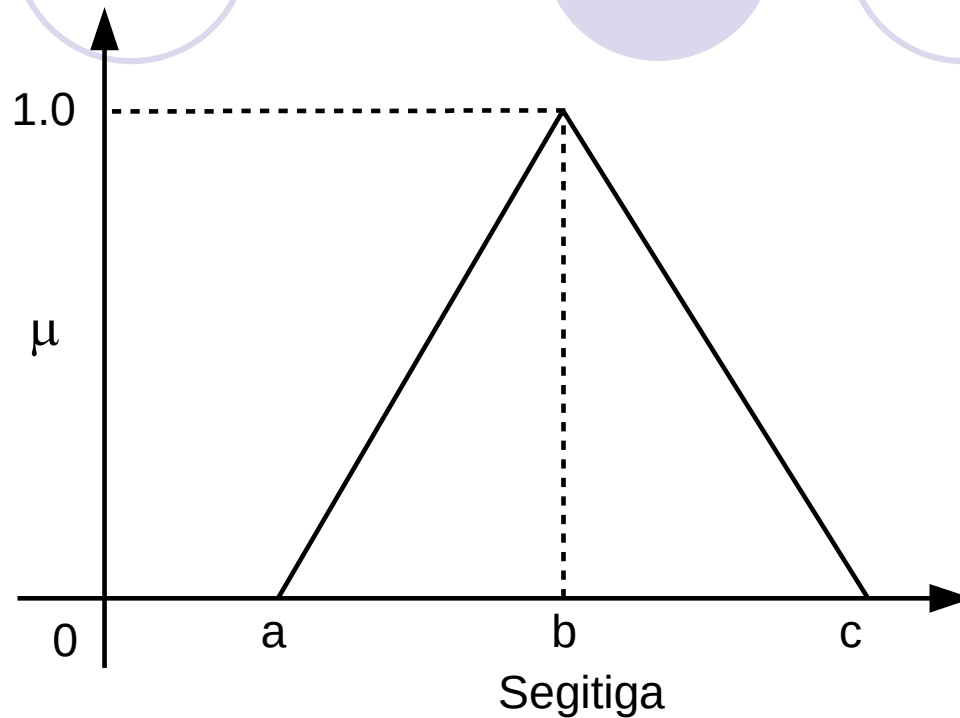
$$\mu[x] = \begin{cases} 0; & x \leq a \\ (x-a)/(b-a); & a < x \leq b \\ 1; & x > b \end{cases}$$



Linier Turun

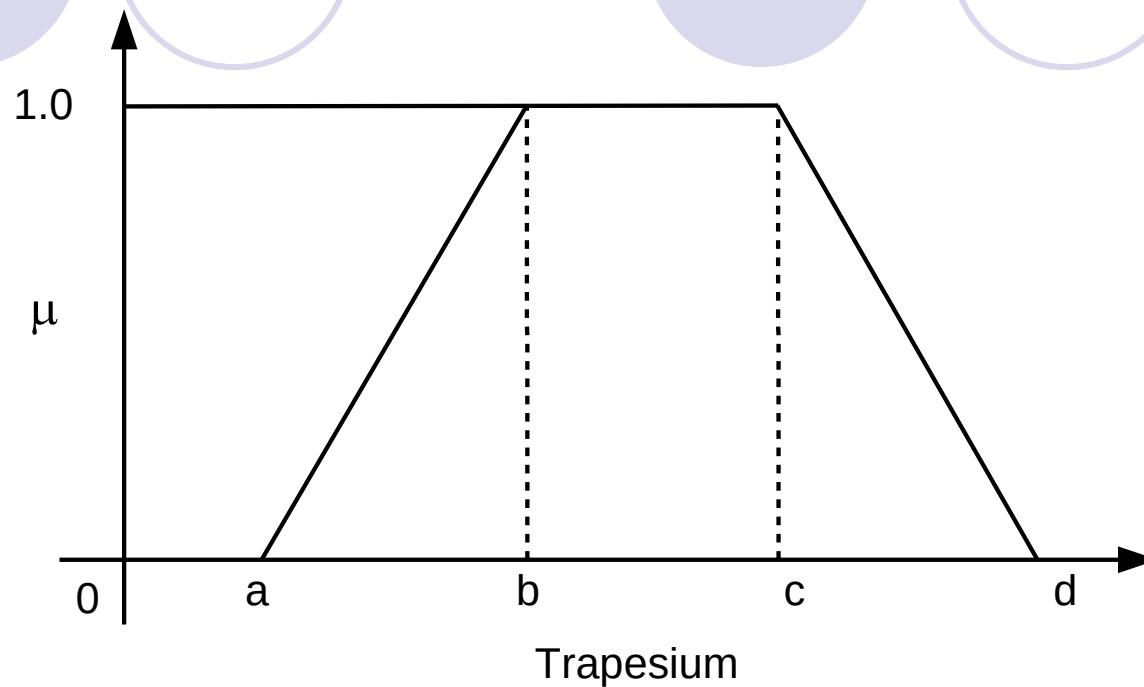
$$\mu[x] = \begin{cases} (b-x)/(b-a); & a \leq x < b \\ 0; & x \geq b \end{cases}$$

Fungsi Keanggotaan: Segitiga



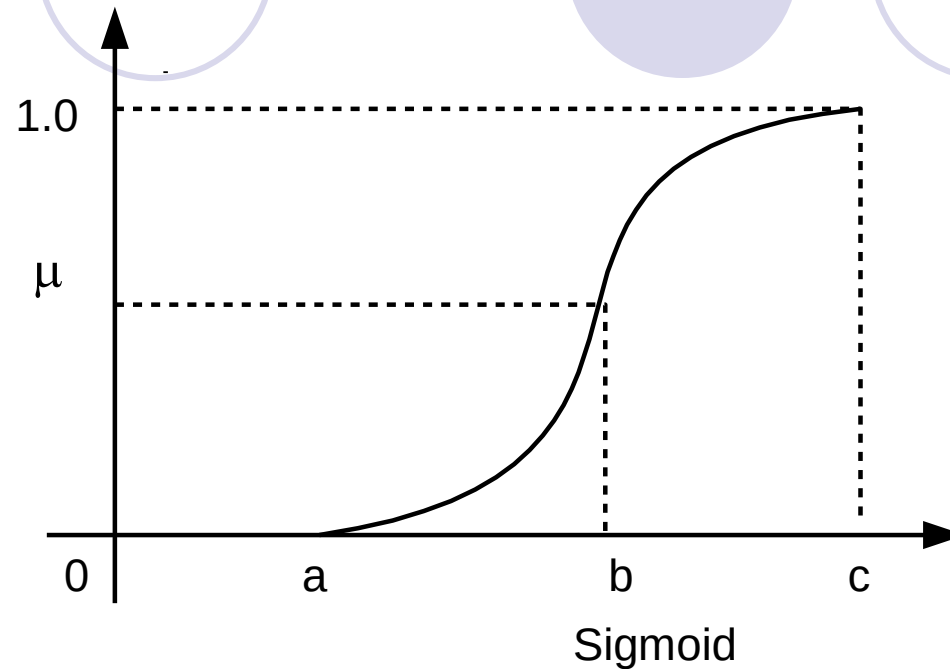
$$\begin{aligned}\mu[x] &= 0; \quad x \leq a \text{ atau } x \geq c \\ &= (x-a)/(b-a); \quad a < x \leq b \\ &= (c-x)/(c-b); \quad b < x < c\end{aligned}$$

Fungsi Keanggotaan: Trapezium



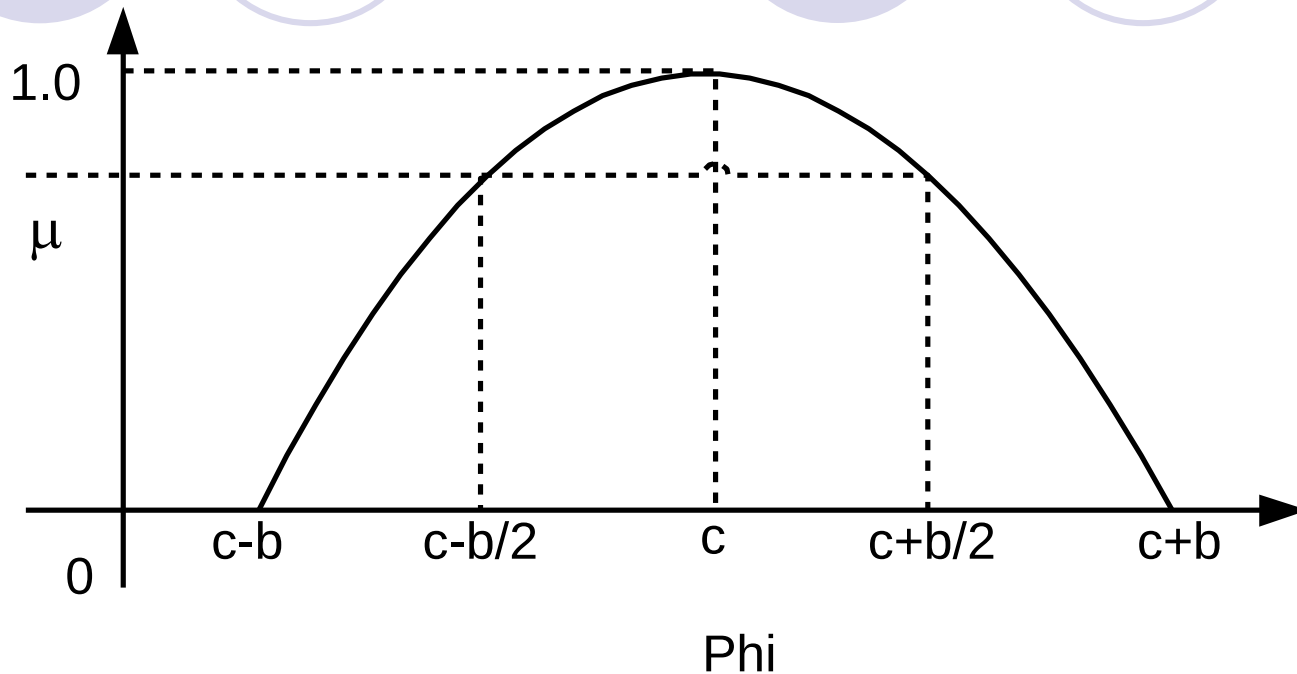
$$\begin{aligned}\mu[x] = & 0; \quad x \leq a \text{ atau } x \geq d \\ & (x-a)/(b-a); \quad a < x \leq b \\ & 1; \quad b < x \leq c \\ & (d-x)/(d-c); \quad c < x < d\end{aligned}$$

Fungsi Keanggotaan: Sigmoid



$$\mu[x;a,b,c]_{\text{sigmoid}} = \begin{cases} 0; & x \leq a \\ 2 \left(\frac{x - a}{c - a} \right)^2; & a < x \leq b \\ 1 - 2 \left(\frac{c - x}{c - a} \right)^2; & b < x < c \\ 1; & x \geq c \end{cases}$$

Fungsi Keanggotaan: Phi



$$\mu[x;a,b,c]_{\text{phi}} = \mu[x;c-b,c-b/2,c]_{\text{sigmoid}}; \quad x \leq c$$

$$\mu[x;c,c+b/2,c+b]_{\text{sigmoid}}; \quad x > c$$

Fuzzy Rules

- In 1973, Lotfi Zadeh published his second most influential paper. This paper outlined a new approach to analysis of complex systems, in which Zadeh suggested capturing human knowledge in **fuzzy rules**.
- A fuzzy rule can be defined as a conditional statement in the form:

IF	x	is A
THEN	y	is B
- where x and y are linguistic variables; and A and B are linguistic values determined by fuzzy sets on the universe of discourses X and Y , respectively.

Classical Vs Fuzzy Rules

- A classical IF-THEN rule uses binary logic, for example,

Rule: 1

IF speed is > 100
THEN stopping_distance is long

Rule: 2

IF speed is < 40
THEN stopping_distance is short

- The variable speed can have any numerical value between 0 and 220 km/h, but the linguistic variable stopping_distance can take either value long or short.
- In other words, classical rules are expressed in the black-and-white language of Boolean logic.

Classical Vs Fuzzy Rules

- We can also represent the stopping distance rules in a fuzzy form:

Rule: 1

IF speed is fast
THEN stopping_distance is long

Rule: 2

IF speed is slow
THEN stopping_distance is short

- In fuzzy rules, the linguistic variable speed also has the range (the universe of discourse) between 0 and 220 km/h, but this range includes fuzzy sets, such as slow, medium and fast.
- The universe of discourse of the linguistic variable stopping_distance can be between 0 and 300 m and may include such fuzzy sets as short, medium and long.

Tahapan Fuzzy Inference Rule

- **Fuzzification:** definisi dari himpunan fuzzy dan penentuan derajat keanggotaan dari *crisp input* pada sebuah himpunan fuzzy
- **Inferensi:** evaluasi kaidah/aturan/rule fuzzy untuk menghasilkan output dari tiap rule
- **Composisi:** agregasi atau kombinasi dari keluaran semua rule
- **Defuzzification:** perhitungan *crisp output*

Model Fuzzy Mamdani

Contoh: persoalan sederhana dengan 2 input (x dan y), 1 output (z) dan 3 rules

Rule: 1

IF x is A3
OR y is B1
THEN z is C1

Rule: 2

IF x is A2
AND y is B2
THEN z is C2

Rule: 3

IF x is A1
THEN z is C3

Rule: 1

IF project_funding is adequate
OR project_staffing is small
THEN risk is low

Rule: 2

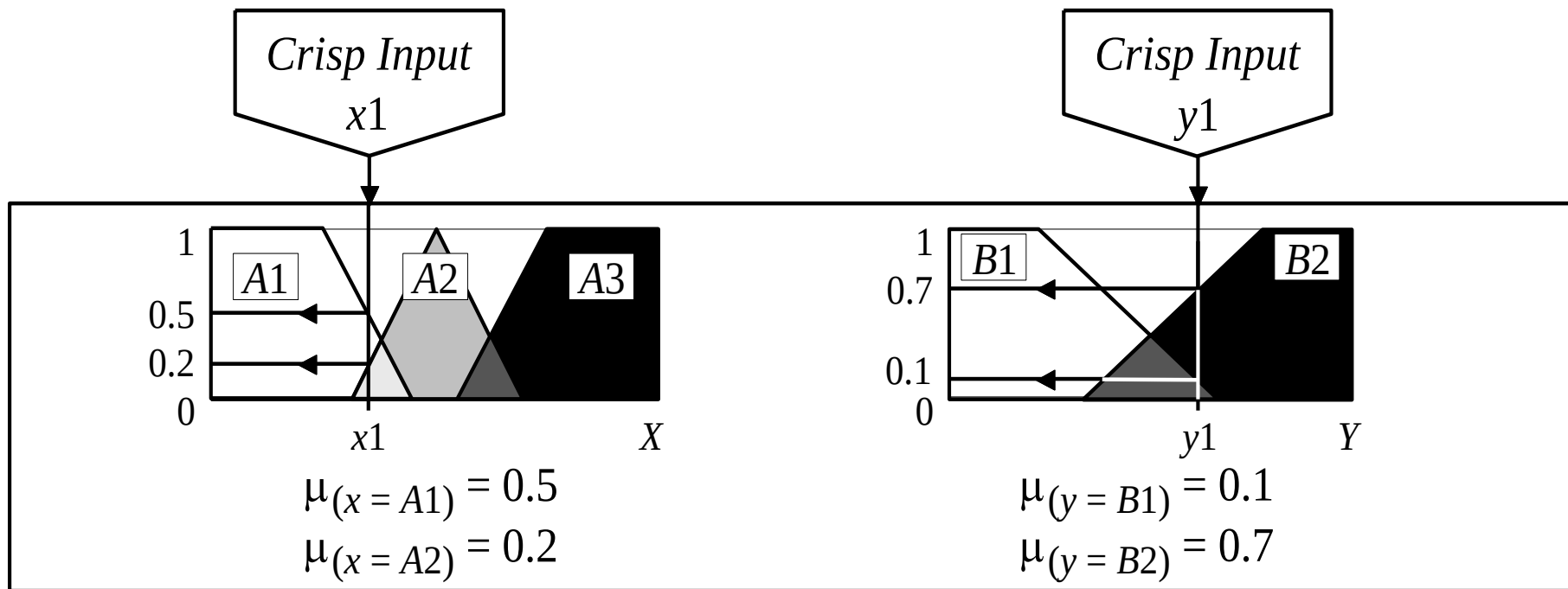
IF project_funding is marginal
AND project_staffing is large
THEN risk is normal

Rule: 3

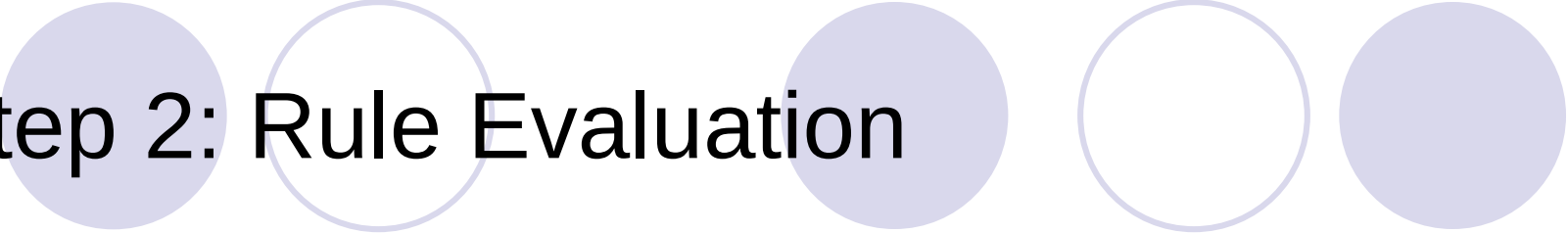
IF project_funding is inadequate
THEN risk is high

Step 1: Fuzzification

Fuzzifikasi: menentukan derajat keanggotaan input $x1$ dan $y1$ pada himpunan fuzzy



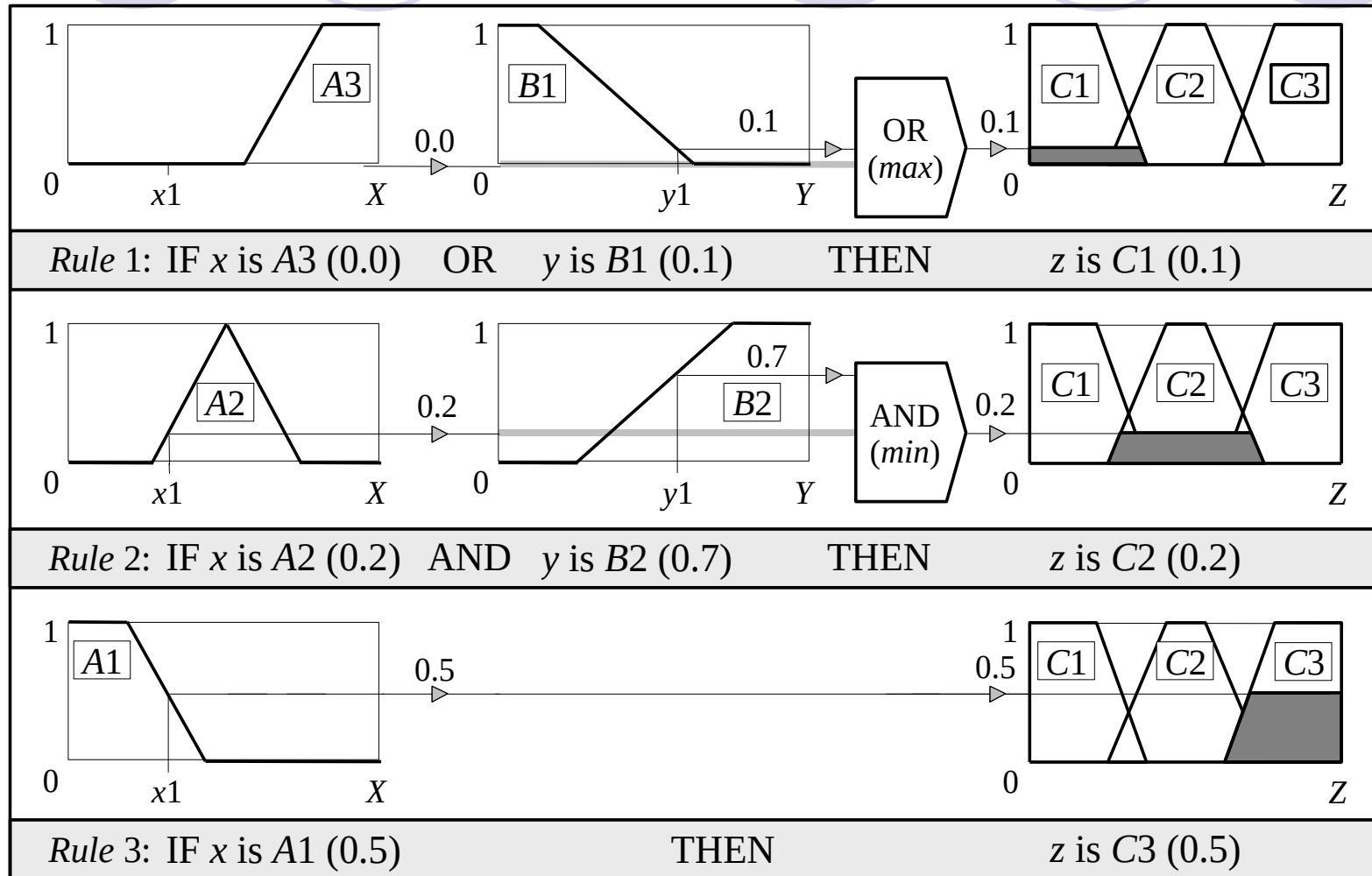
Step 2: Rule Evaluation



Inferensi: aplikasikan fuzzified inputs, $\mu(x=A1) = 0.5$, $\mu(x=A2) = 0.2$, $\mu(y=B1) = 0.1$ and $\mu(y=B2) = 0.7$, ke anteseden dari aturan fuzzy

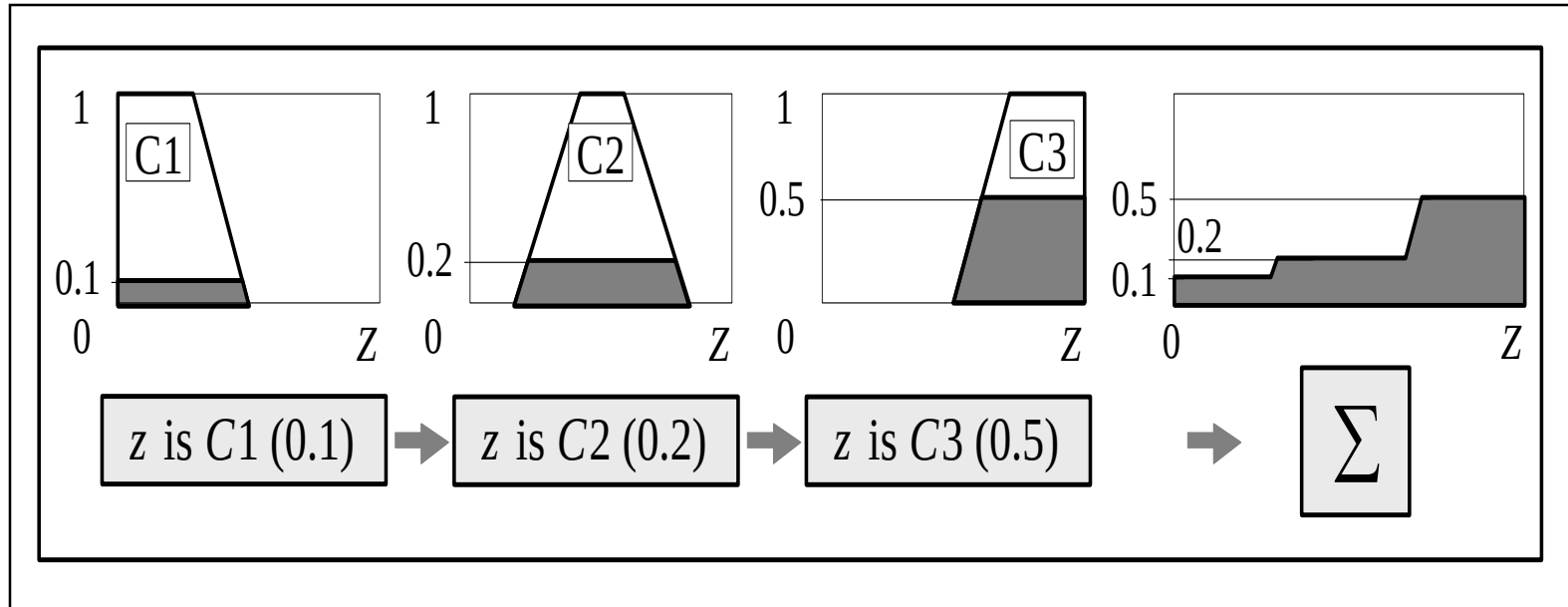
Untuk aturan fuzzy dengan anteseden lebih dari 1, operator fuzzy (AND atau OR) digunakan untuk mencapai sebuah nilai tunggal yang merepresentasikan hasil rule fuzzy. Nilai ini kemudian diaplikasikan ke fungsi keanggotaan konsekuen

Step 2: Rule Evaluation



Step 3: Aggregation of the rule outputs

Agregasi keluaran semua rule ke dalam himpunan fuzzy tunggal.



Step 4: Defuzzification

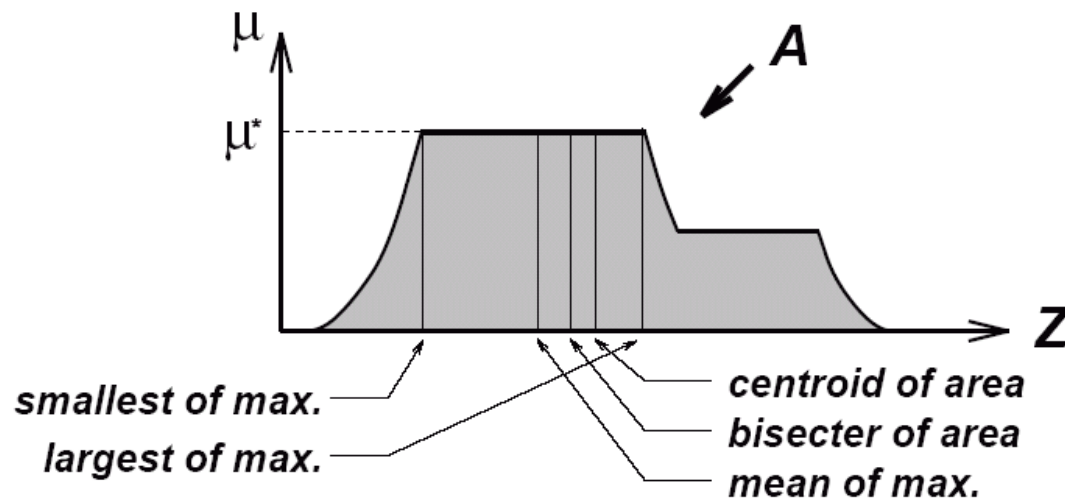
Defuzzifikasi: konversi dari himpunan fuzzy yang dihasilkan dari komposisi ke dalam crisp value.

Teknik yang paling populer adalah **centroid technique**. Metoda ini mencari **centre of gravity (COG)** dari *aggregate set*:

$$COG = \frac{\int_a^b \mu_A(x) x dx}{\int_a^b \mu_A(x) dx}$$

Five commonly used defuzzifying methods:

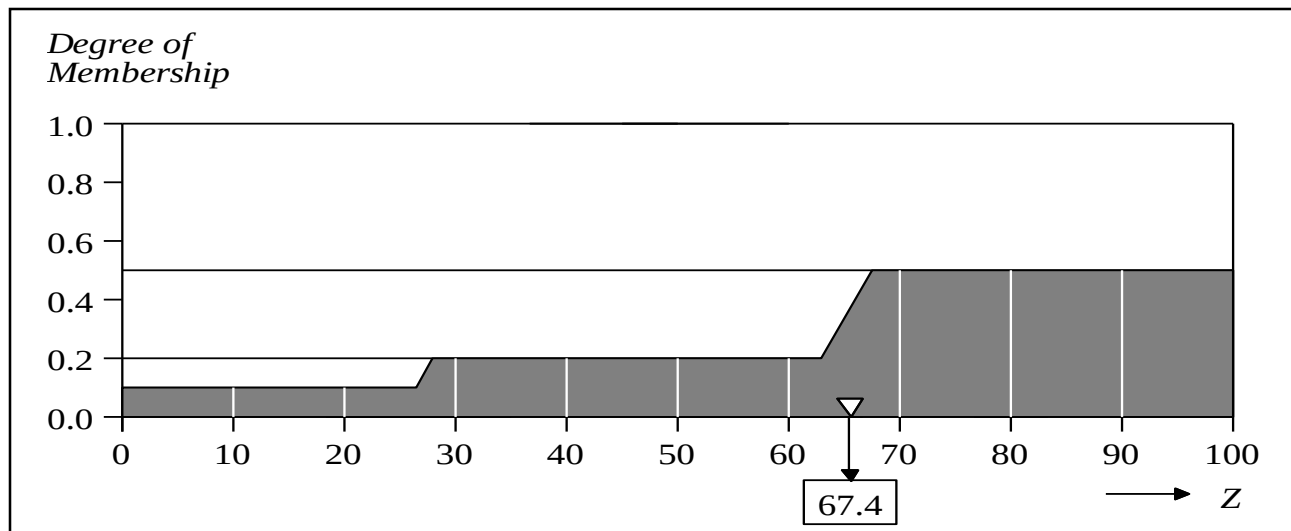
- Centroid of area (COA)
- Bisector of area (BOA)
- Mean of maximum (MOM)
- Smallest of maximum (SOM)
- Largest of maximum (LOM)



Model Fuzzy Mamdani

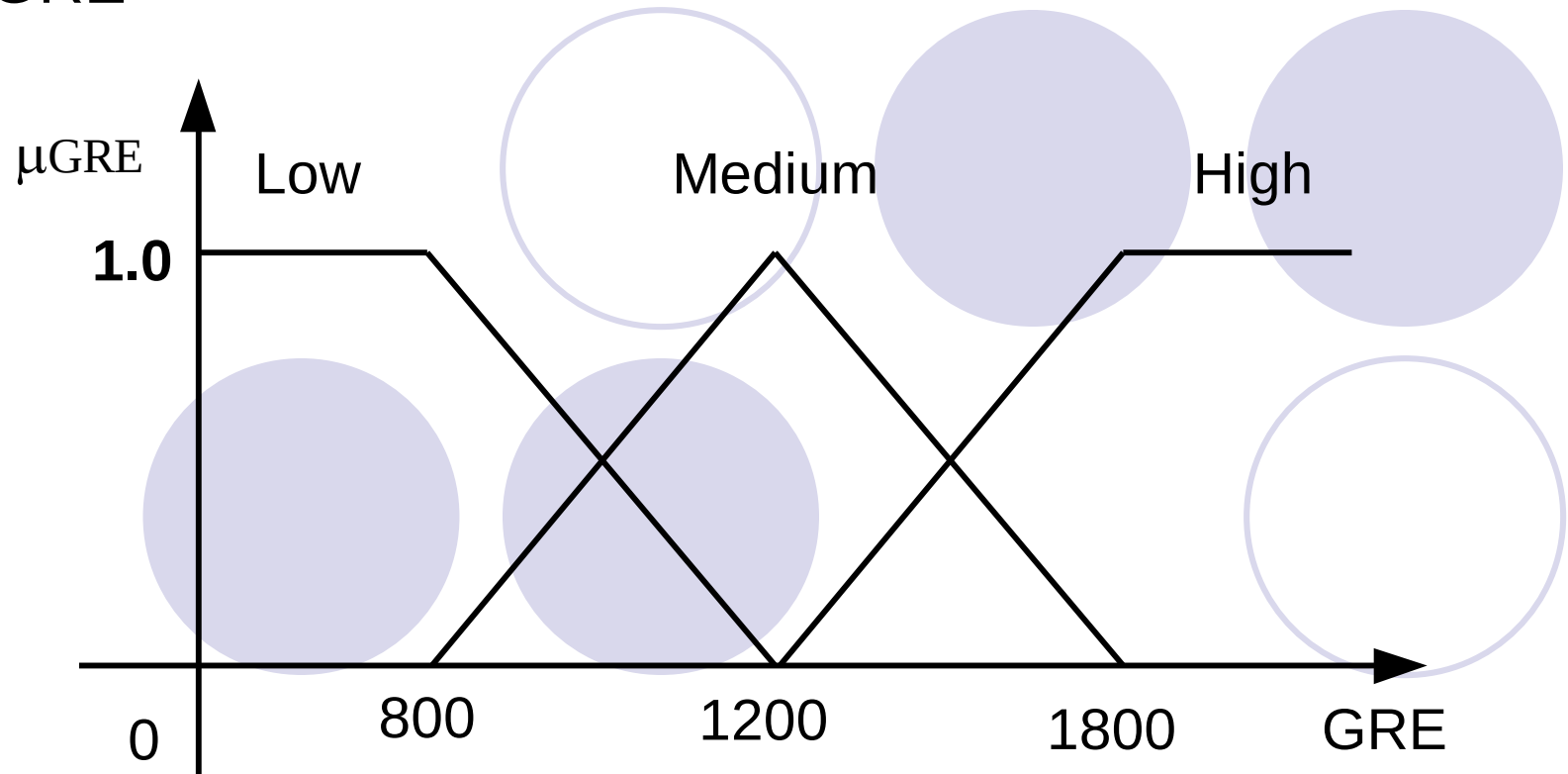
Centre of gravity (COG): mencari titik yang membagi area solusi menjadi 2 bagian yang sama

$$COG = \frac{(0+10+20) \times 0.1 + (30+40+50+60) \times 0.2 + (70+80+90+100) \times 0.5}{0.1+0.1+0.1+0.2+0.2+0.2+0.2+0.5+0.5+0.5+0.5} = 67.4$$



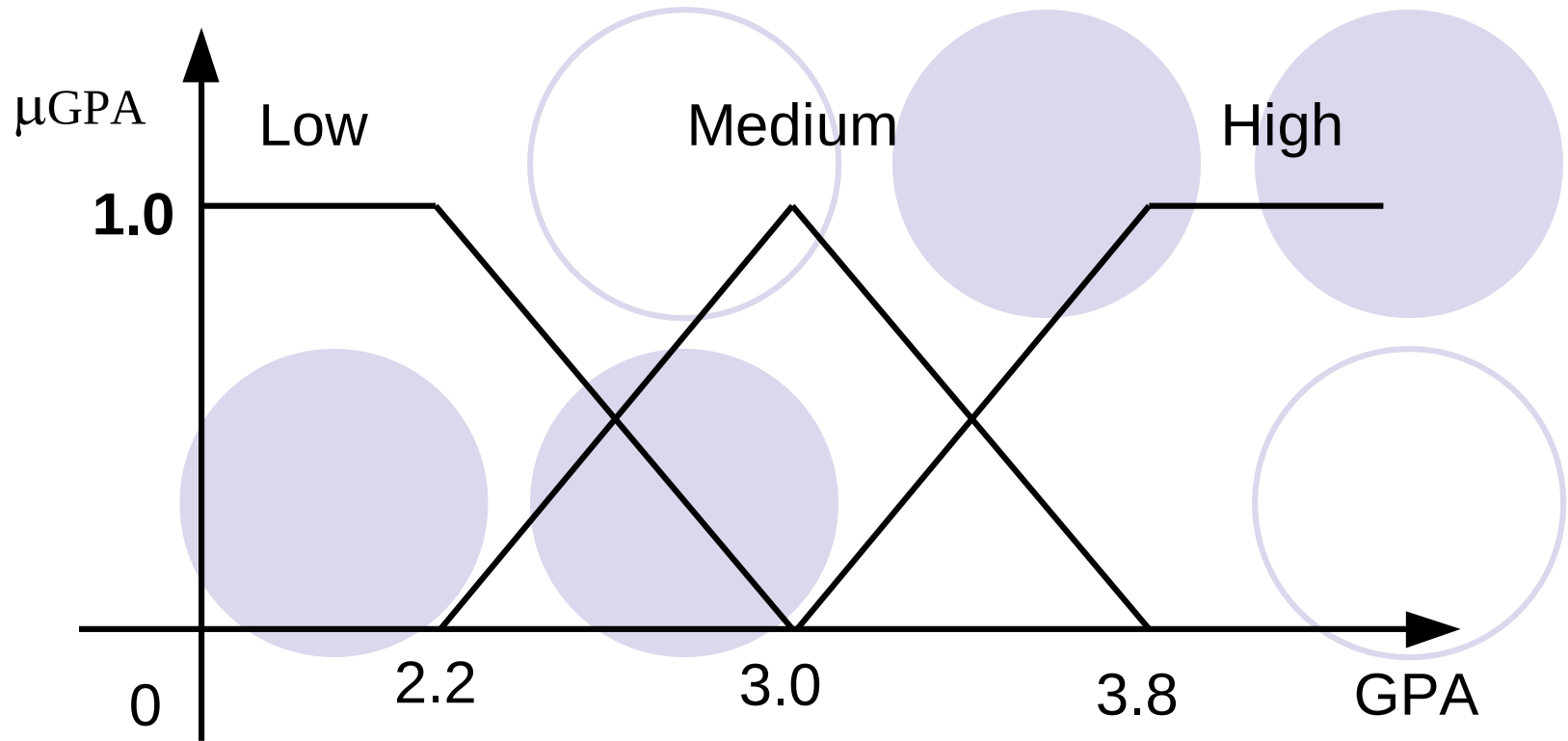
Latihan

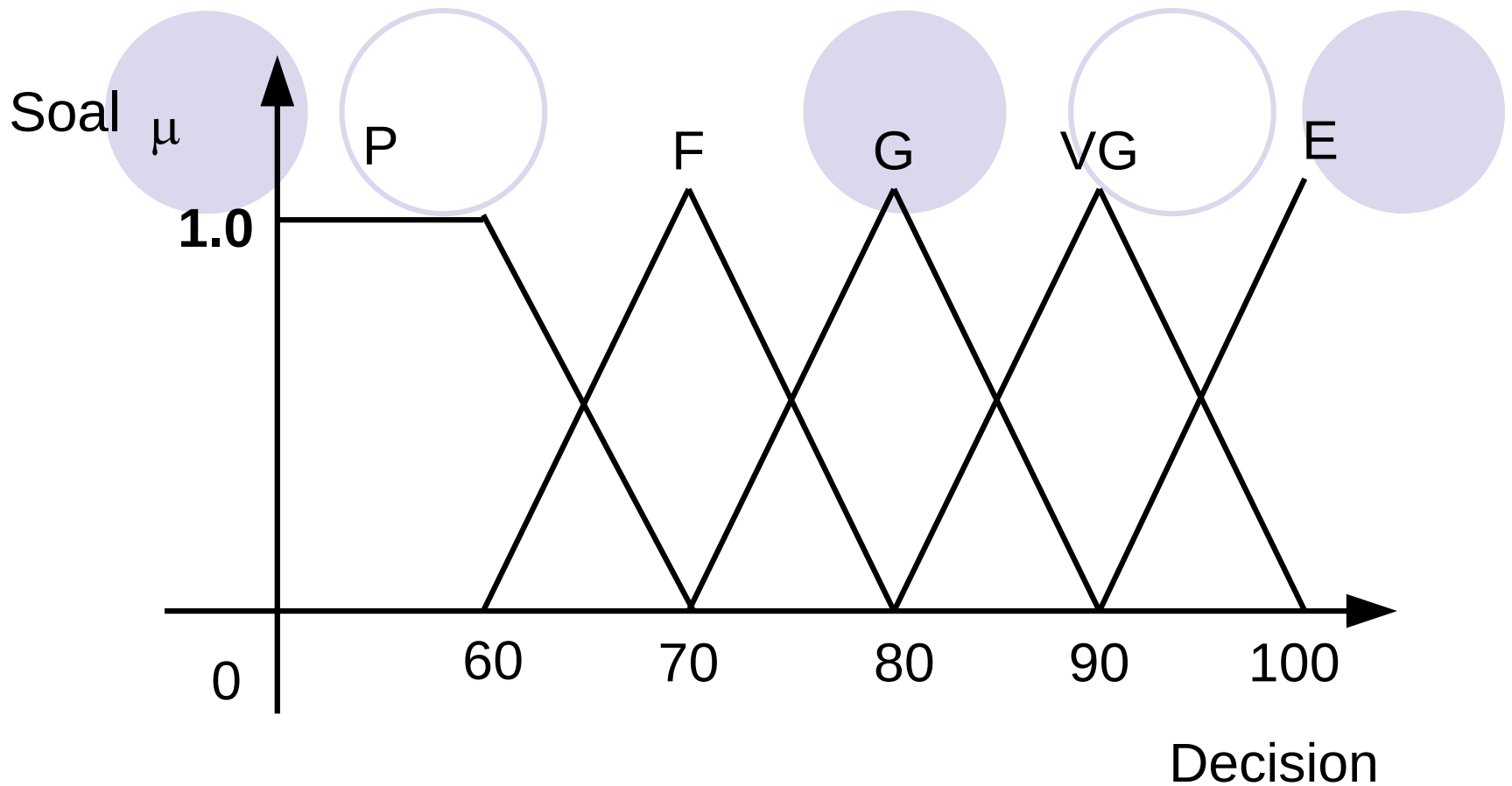
Mengevaluasi mahasiswa berdasarkan GPA dan nilai GRE



Fungsi Keanggotaan untuk GRE

Fungsi Keanggotaan untuk GPA





- E = Excellent
- VG = Very Good
- G = Good
- F = Fair
- P = Poor

GRE				
G P A		H	M	L
	H	E	VG	F
	M	G	G	P
	L	F	P	P

H = High
M = Medium
L = Low

E = Excellent
VG = Very Good
G = Good

F = Fair
P = Poor

Model Fuzzy Sugeno

Inferensi Mamdani tidak efisien karena melibatkan proses pencarian centroid dari area 2 dimensi.

Michio Sugeno mengusulkan penggunaan singleton sebagai fungsi keanggotaan dari konsekuen.

Singleton adalah sebuah himpunan fuzzy dengan fungsi keanggotaan: pada titik tertentu mempunyai sebuah nilai dan 0 di luar titik tersebut.

Model Fuzzy Sugeno

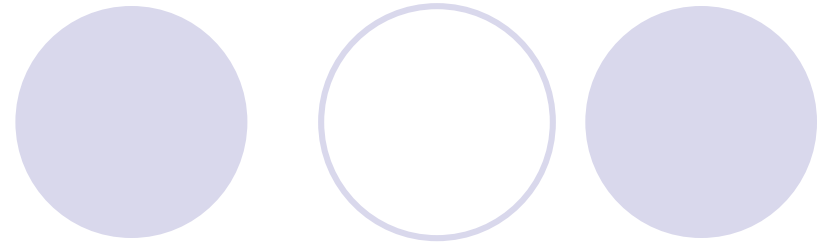
Perbedaan antara Mamdani dan Sugeno ada pada konsekuensi. Sugeno menggunakan konstanta atau fungsi matematika dari variabel input:

IF x is A
AND y is B
THEN z is $f(x, y)$

IF x is A
AND y is B
THEN z is k

dimana x , y dan z adalah variabel linguistik; A dan B himpunan fuzzy untuk X dan Y, dan $f(x, y)$ adalah fungsi matematik.

Examples



R1: if X is small and Y is small then $z = -x + y + 1$

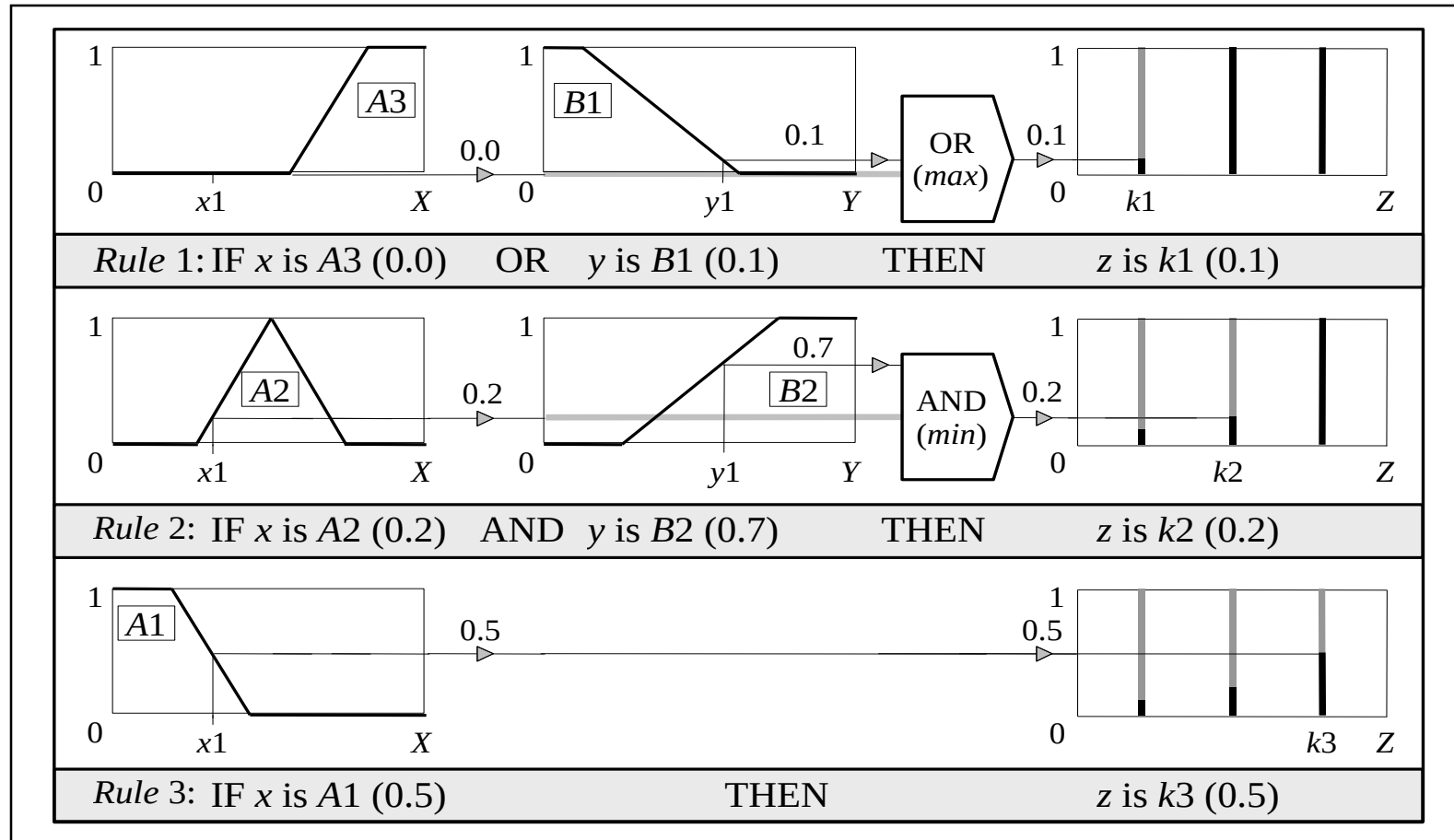
R2: if X is small and Y is large then $z = -y + 3$

R3: if X is large and Y is small then $z = -x + 3$

R4: if X is large and Y is large then $z = x + y + 2$

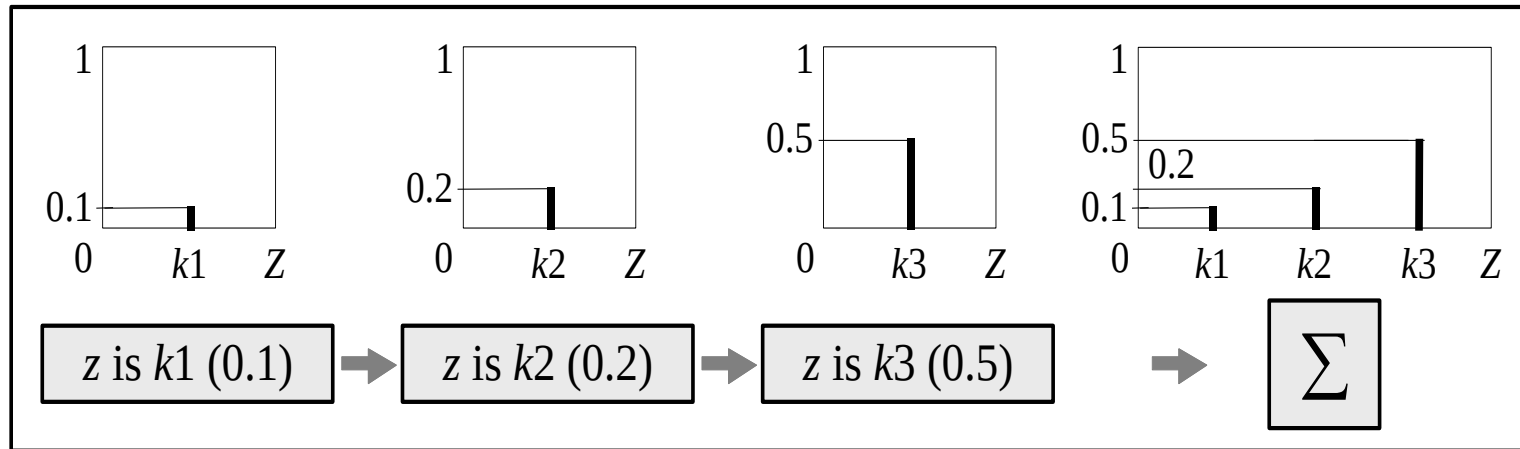
Model Fuzzy Sugeno

Step 2: Rule Evaluation



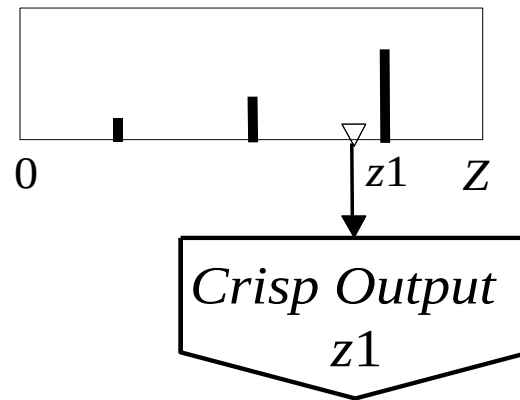
Model Fuzzy Sugeno

Step 3: Aggregation of the rule outputs



Model Fuzzy Sugeno

Defuzzifikasi



Weighted average (WA):

$$WA = \frac{\mu(k1) \times k1 + \mu(k2) \times k2 + \mu(k3) \times k3}{\mu(k1) + \mu(k2) + \mu(k3)} = \frac{0.1 \times 20 + 0.2 \times 50 + 0.5 \times 80}{0.1 + 0.2 + 0.5} = 65$$

Model Fuzzy Sugeno: Contoh

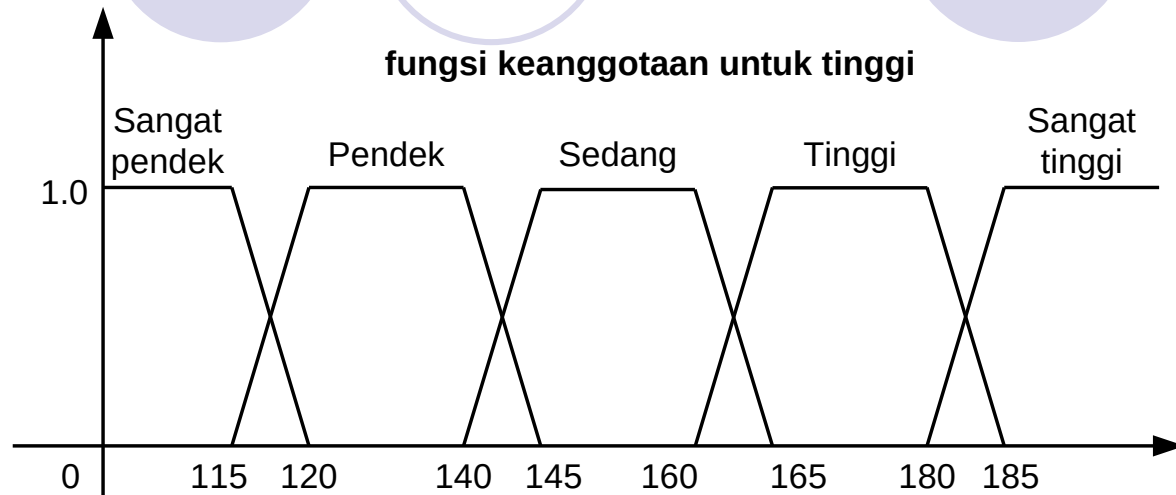
Mengevaluasi kesehatan orang berdasarkan tinggi dan berat badannya

Input: tinggi dan berat badan

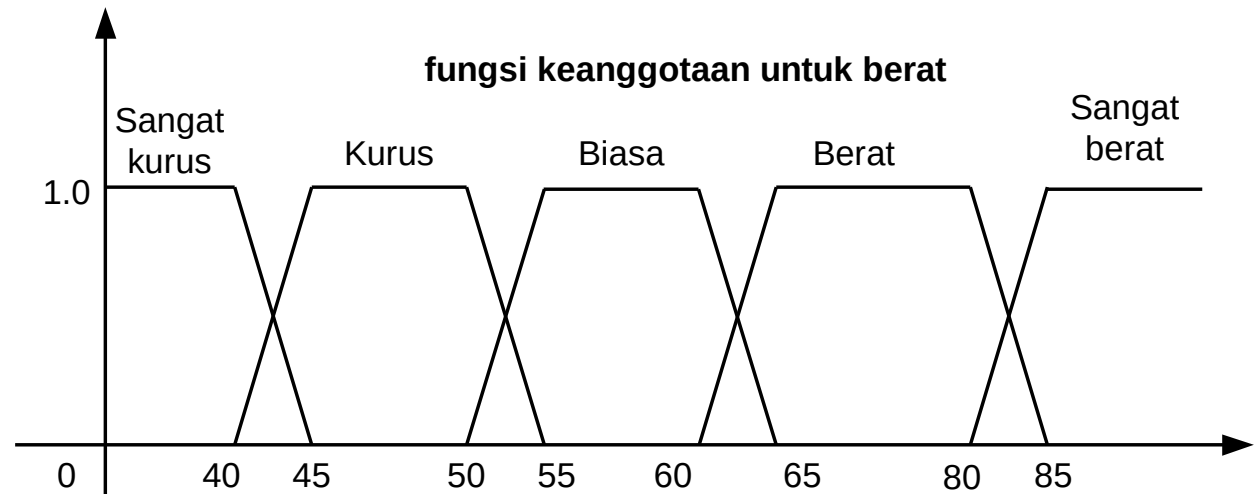
Output: kategori sehat

- sangat sehat (SS), index=0.8
- sehat (S), index=0.6
- agak sehat (AS), index=0.4
- tidak sehat (TS), index=0.2

L1: Fuzzification (1)



Ada 3 variabel fuzzy yang dimodelkan: tinggi, berat, sehat



L2: Rules Evaluation (1)

Tentukan rules

Tabel Kaidah Fuzzy

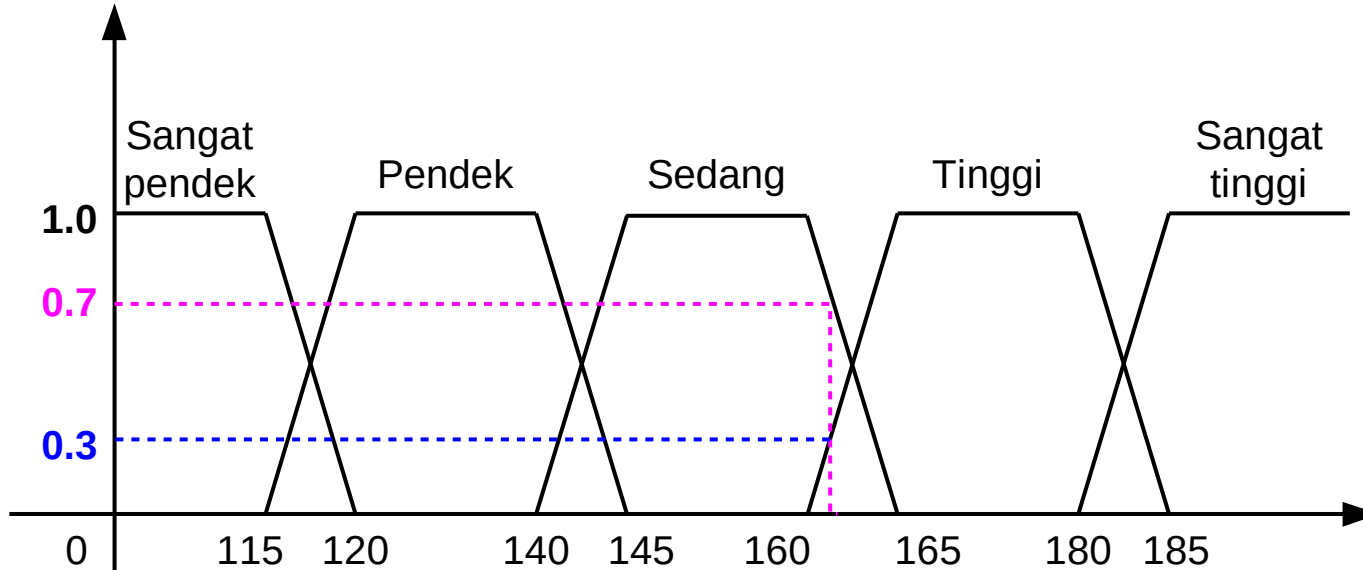
BERAT						
T I N G G I		Sangat kurus	Kurus	Biasa	Berat	Sangat berat
	Sangat pendek	SS	S	AS	TS	TS
	Pendek	S	SS	S	AS	TS
	Sedang	AS	SS	SS	AS	TS
	Tinggi	TS	S	SS	S	TS
	Sangat tinggi	TS	AS	SS	S	AS

Dalam bentuk if-then, contoh:

*If sangat pendek dan sangat kurus then
sangat sehat*

L2: Rules Evaluation (2)

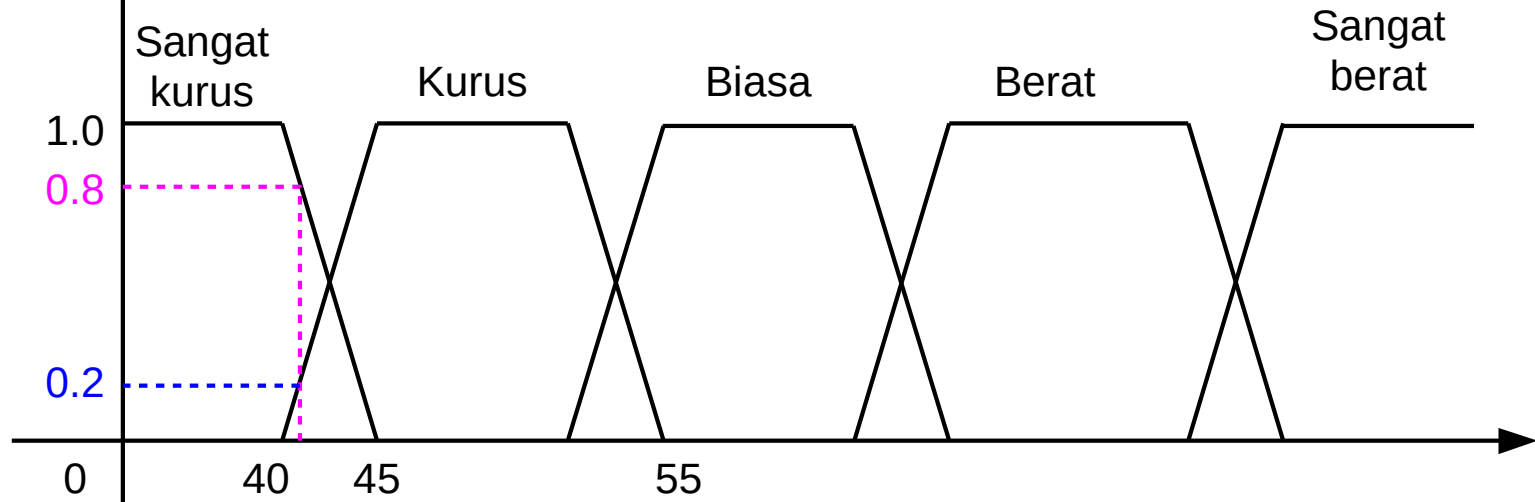
Contoh: bagaimana kondisi kesehatan untuk orang dengan tinggi 161.5 cm dan berat 41 kg?



$$\mu_{\text{sedang}}[161.5] = (165 - 161.5) / (165 - 160) = 0.7$$

$$\mu_{\text{tinggi}}[161.5] = (161.5 - 160) / (165 - 160) = 0.3$$

L2: Rules Evaluation (3)



$$\mu_{\text{sangat kurus}}[41] = (45 - 41) / (45 - 40) = 0.8$$

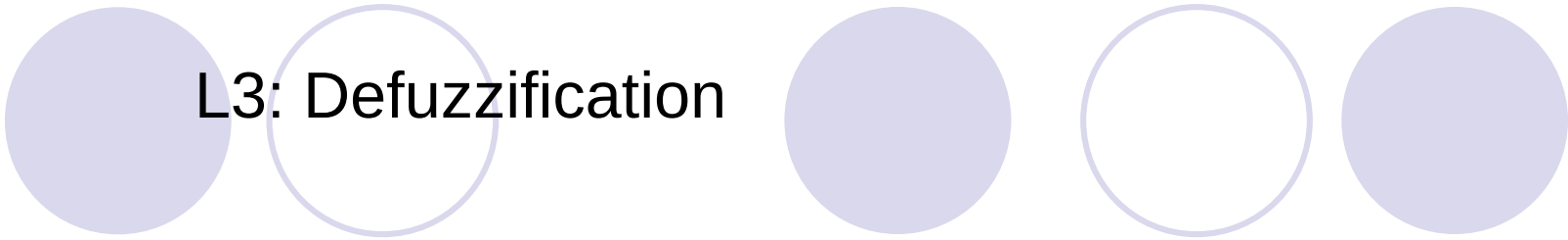
$$\mu_{\text{kurus}}[41] = (41 - 40) / (45 - 40) = 0.2$$

BERAT						
T I N G G I		0.8	0.2	Biasa	Berat	Sangat berat
	Sangat pendek	SS	S	AS	TS	TS
	Pendek	S	SS	S	AS	TS
	0.7	AS	SS	SS	AS	TS
	0.3	TS	S	SS	S	TS
	Sangat tinggi	TS	AS	SS	S	AS

L2: Rules Evaluation (4)

Pilih bobot minimum
krn relasi AND

BERAT						
T I N G G I		0.8	0.2	Biasa	Berat	Sangat berat
	Sangat pendek	SS	S	AS	TS	TS
	Pendek	S	SS	S	AS	TS
	0.7	0.7	0.2	SS	AS	TS
	0.3	0.3	0.2	SS	S	TS
	Sangat tinggi	TS	AS	SS	S	AS



L3: Defuzzification

Diperoleh:

$$f = \{TS, AS, S, SS\} = \{0.3, 0.7, 0.2, 0.2\}$$

Penentuan hasil akhir, ada 2 metoda:

1. Max method: index tertinggi 0.7
hasil Agak Sehat

2. Centroid method, dengan metoda **Sugeno**:

$$\begin{aligned}\text{Decision Index} &= (0.3 \times 0.2) + (0.7 \times 0.4) + (0.2 \times 0.6) + (0.2 \times 0.8) / \\ &\quad (0.3 + 0.7 + 0.2 + 0.2) \\ &= 0.4429\end{aligned}$$

Crisp decision index = 0.4429

Fuzzy decision index: 75% agak sehat, 25% sehat

Summary



- Ada 4 tahapan utama sistem pakar fuzzy: fuzzifikasi, inferensi, komposisi, defuzzifikasi.
- 2 metoda yang paling banyak dipakai: Mamdani dan Sugeno.
- Metoda Mamdani menggunakan himpunan fuzzy sebagai konsekuen rule, Metoda Sugeno menggunakan fungsi matematik atau konstanta.
- Mamdani: komputasi lebih berat, human-like inference, Sugeno: komputasi lebih efisien tetapi kehilangan interpretabilitas linguistik.